

VMEM- φ : LOW-COMPUTE OUT-OF-DISTRIBUTION DETECTION IN SPIKING NEURAL NETWORKS FROM MEMBRANE-POTENTIAL STATISTICS

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ABSTRACT

Spiking neural networks (SNNs) processing event-camera data operate at low power, but standard out-of-distribution (OOD) detection methods rely on floating-point logits or gradients that SNNs lack. We show that the sub-threshold membrane potential $V_{\text{mem}}(t)$ of a Parametric Leaky-Integrate-and-Fire (PLIF) neuron, summarized by its per-channel statistics $\varphi = [\mu, \sigma^2, \kappa]$, gives a nearly zero-cost OOD signal requiring no additional forward passes. To evaluate this, we release **Gen1-C**, a reproducible event-camera corruption benchmark featuring six sensor-grounded corruptions across five severities, alongside a pre-computed membrane-statistics feature set. A closed-form theory of the leaky integrator predicts which corruptions shift specific membrane statistics, and these corruptions indeed sharply degrade the detection accuracy (mAP) of a pretrained hybrid SNN-ANN. Finally, we introduce the Manifold-Decomposition Detector (MDD), an unsupervised, corruption-blind method that scores complementary directions in feature space. MDD achieves ≥ 0.85 AUROC on four of the six corruptions at the highest severity using a bounded 64-frame window; for the remaining two, low detectability tracks low actual damage to the network, the same theory’s prediction for a benign invariance.

1 INTRODUCTION

Event cameras record per-pixel brightness changes as they happen, at microsecond resolution, with high dynamic range and low power draw. Paired with spiking neural networks (SNNs), which also run at low power, they make an efficient sensing-and-processing pipeline for safety-critical uses (self-driving cars, robotics), where one question matters most: *if the camera’s input drifts away from what the network was trained on, can the system tell?* Out-of-distribution (OOD) detection is the usual answer for ordinary (non-spiking) neural networks, but nearly every method in that literature needs something a low-power spiking network does not cheaply produce: floating-point class scores and a softmax (MSP, Energy), floating-point internal features (ViM, ReAct), input gradients or a backward pass (ODIN, GradNorm), or a separate, non-spiking feature network (ANN-feature Mahalanobis/kNN), none of which a spiking network produces on its own (Section C).

We start from a signal that costs almost nothing to obtain. A Parametric Leaky-Integrate-and-Fire (PLIF) neuron keeps a sub-threshold membrane potential $V_{\text{mem}}(t)$, an internal value the network already updates at every time step to decide whether to fire. Reading this value and summarizing it with a few per-channel statistics adds no extra forward-pass computation. We call this summary φ and ask how good a cheap signal it makes for detecting realistic event-camera corruptions, each grounded in an actual sensor failure mechanism (Figure 1).

Contributions.

- **A low-compute membrane OOD signal and the Gen1-C benchmark.** We introduce φ , a summary of the PLIF membrane state, and release **Gen1-C**: a reproducible (seeded, model-agnostic) event-camera corruption benchmark with six sensor-grounded corruptions, each at five

increasing severities (Section 3.2, Figure 1), in the same spirit as ImageNet-C (Hendrycks & Dietterich, 2019), along with the membrane-statistics feature set we extracted.

- **A closed-form theory of detectability, and why detectability tracks harm.** We derive, from the PLIF state equation with no fitted parameters, a taxonomy of which corruptions move which membrane moments and prove severity monotonicity with an exact bound on the reset nonlinearity (Section 3.3).
- **The Manifold-Decomposition Detector (MDD).** MDD scores several complementary, two-sided directions in feature space and combines them with a calibrated OR rule (Section 3.5), reaching ≥ 0.85 AUROC on four of the six corruptions at the highest severity using a bounded 64-frame window (Section 4.8).

2 BACKGROUND AND RELATED WORK

OOD detection for ANNs. The dominant families score a post-hoc statistic of a trained network – softmax/logit-based, activation- or feature-space distance, and gradient-based methods, surveyed under the generalised-OOD taxonomy we follow (Yang et al., 2024) and tabulated by name against their SNN-incompatible requirements in Table 5 (Appendix C). Most target classifiers rather than detectors; the closest to our setting is VOS (Du et al., 2022), which synthesises virtual outliers to regularise an ANN detection head.

OOD detection for SNNs. Prior SNN-OOD methods work on image *classification* and read the *spikes*: Martínez-Seras et al. (Martínez-Seras et al., 2023) score the typicality of per-layer spike-count patterns; Terres-Escudero et al. (Terres-Escudero et al., 2024) use the sparse latent representations of Forward-Forward-trained spiking networks; and Avramović et al. (Avramović et al., 2025a;b) measure distances between spike patterns and class representatives in (residual) SNNs. We instead read the *sub-threshold membrane potential* of an event-camera *object detector*, against the corruptions we release as Gen1-C.

Event-camera robustness and corruption benchmarks. Robustness benchmarks built from synthetic corruptions are standard for frame cameras (ImageNet-C (Hendrycks & Dietterich, 2019)). The Prophesee Gen1 automotive dataset (de Tournemire et al., 2020) (39+ hours of driving, 255k+ bounding-box labels) provides our evaluation data. N-ImageNet (Kim et al., 2021) established robustness testing for event-based classifiers by physically re-recording under varied camera trajectories and lighting. Hu et al. (Hu et al., 2021) simulate realistic DVS noise (threshold mismatch, finite bandwidth, shot noise) from video frames; Gen1-C instead applies synthetic, severity-graded, sensor-fault corruptions directly to histogram tensors.

Host detector. We read φ read-only, never retraining, from the pretrained hybrid SNN-ANN detector of Ahmed et al. (2025), which fuses a spiking front-end (SpikingJelly PLIF (Fang et al., 2021)) with an ANN detection head.

3 METHOD

3.1 THE MEMBRANE DESCRIPTOR φ

We read the membrane signal from the pretrained hybrid SNN-ANN detector using a forward hook attached to all four PLIF layers (704 channels total). Concretely, V is the neuron’s membrane state at each of the $T=10$ unrolled time steps; we read the raw membrane state, including the steps right after a spike resets it, rather than removing them. Because of this, the linear, sub-threshold analysis in Section 3.3 is an approximation, and the effect of the reset enters as a bounded error term (Theorem 1) that vanishes when firing is rare. For each channel c we apply global average pooling (GAP) over the spatial dimensions of V , then compute its first three central moments across the T time steps,

$$\varphi_c = [\mu_c, \sigma_c^2, \kappa_c], \quad \mu_c = \mathbb{E}[V_c], \quad \sigma_c^2 = \text{Var}[V_c], \quad \kappa_c = \frac{\mathbb{E}[(V_c - \mu_c)^4]}{\sigma_c^4}, \quad (1)$$

which stack into a 2112-D descriptor φ per frame. The mean captures sustained drive (Corollary 1) and the variance captures fluctuation energy (Corollary 2); the kurtosis is sensitive to non-Gaussian,

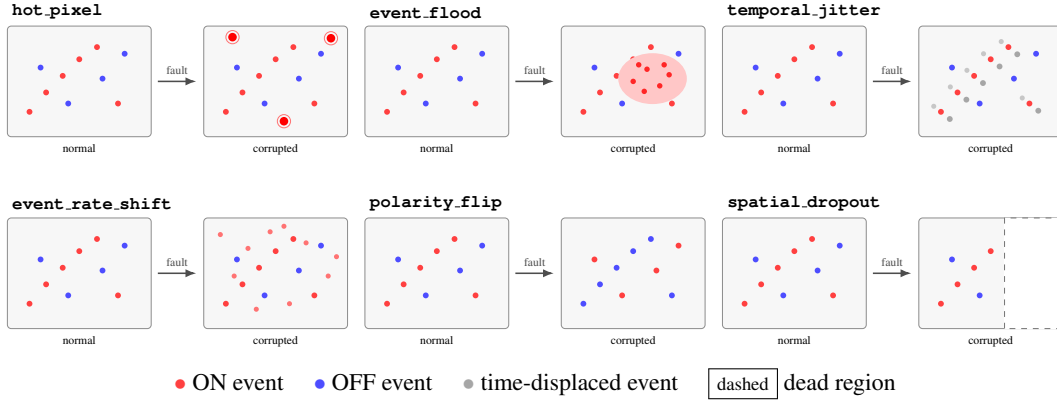


Figure 1: **The six Gen1-C corruptions.** For each: a normal event frame (left) and the corrupted frame the modelled hardware fault produces (right). Mechanisms are described in Section 3.2.1; the induced membrane signatures in Section 3.3.

bursty fluctuations. (i) GAP throws away spatial layout, so φ knows *how much* each channel fired but not *where*; (ii) φ can inherit invariances of the trained network – we show this formally for one case, contingent on the network being approximately polarity-symmetric, in Appendix B.3 (Proposition 2). A spatial-dispersion extension, `phi_spatial` (1408-D: per-channel spatial variance and participation ratio), recovers part of the spatial structure that GAP discards.

3.2 THE GEN1-C EVENT-CAMERA CORRUPTION SUITE

We define a controlled, reproducible battery of six corruptions that model the dominant failure modes of event-camera front-ends, each at five monotonically increasing severities; with the clean baseline this yields $6 \times 5 + 1 = 31$ runs. Each perturbs a *structurally distinct* axis of the input – DC offset, localized burst, temporal order, global rate, polarity, or spatial support – which is what makes the closed-form analysis of Section 3.3 possible.

Input representation and design principles. Every corruption operates on the model’s input tensor, so it composes with any downstream architecture and is label-free. A sequence is a stack of N event-histogram frames of shape $(20, H, W)$, $H \times W = 240 \times 304$ for Gen1. Each frame aggregates a 50 ms window of events; the 20 channels are two polarities $\times$ ten 5 ms time bins: channels 0–9 ON (oldest to newest), 10–19 OFF. Every random decision is seeded, so a (corruption, severity, seed) triple is deterministic, and severities are monotone (severity 1 mild to 5 severe, Table 8), enabling the Spearman- ρ analysis of Section 4.

3.2.1 PHYSICAL MECHANISMS: HOW EACH CORRUPTION ARISES IN HARDWARE

Each corruption is the tensor-domain analogue of a *documented* event-camera (DVS) non-ideality: hot-pixel / background-activity noise (Hu et al., 2021; Guo & Delbruck, 2023), LED-flicker and readout-saturation bursts (Wang et al., 2022b; Gehrig & Scaramuzza, 2024), arbiter/timestamp jitter (Gehrig & Scaramuzza, 2024; Gallego et al., 2022), per-pixel contrast-threshold and bias drift (Wang et al., 2019; Graça et al., 2023), ON/OFF threshold asymmetry (Wang et al., 2019; Hu et al., 2021), and dead pixels / occlusion (Gallego et al., 2022). Figure 1 contrasts the normal and corrupted sensor output and names each mechanism; the sensor failure, the perturbed input axis, and the exact mapping from each fault to our histogram operation are detailed in Appendix D (Table 7).

3.2.2 CORRUPTION ALGORITHMS

Each corruption operates on a histogram $X \in \{0, \dots, 255\}^{N \times 20 \times H \times W}$ and returns a corrupted copy:

- **hot_pixel**: add a constant δ to every bin/frame at n fixed random locations (`int16`, clipped).

- **event_flood**: at b random frame/patch centres, add δ inside a patch covering fraction f , over frames within ± 2 of the centre.
- **temporal_jitter**: roll the ON block (ch. 0–9) and OFF block (ch. 10–19) along the time-bin axis by independent integer shifts in $[-s_{\max}, s_{\max}]$, zero-filling vacated bins.
- **event_rate_shift**: multiply all counts by a seeded under/over factor and clip.
- **polarity_flip**: swap the ON/OFF blocks on a random fraction p of frames.
- **spatial_dropout**: zero k rectangular regions of size (f_h, f_w) across all channels/frames.

Severity schedules are in Table 8 (Appendix E).

3.3 WHY MEMBRANE STATISTICS DETECT CORRUPTION: A LEAKY-INTEGRATOR THEORY

The detectability of each corruption is a closed-form consequence of the PLIF state equation. The membrane is a leaky exponential accumulator of the input current; each corruption perturbs that current in a structurally distinct way, and the perturbation propagates to a predictable change in the moments of φ .

State equation and closed form. SpikingJelly’s parametric LIF node (Fang et al., 2021) updates, per channel, $V[t] = (1 - \frac{1}{\tau})V[t-1] + WX[t]$, emitting a spike with reset when V crosses threshold θ . Writing $\lambda \equiv 1 - 1/\tau \in (0, 1)$ and unrolling from $V[-1] = 0$,

$$V[t] = \sum_{k=0}^t \lambda^k W X[t-k], \quad (2)$$

so V is a causal, leaky linear filter of the event stream with memory horizon $\sim \tau$. Two consequences: V carries the temporal structure of recent input (so temporally-structured corruptions leave a signature even when the marginal input is unchanged), and because $X \rightarrow V$ is linear, the moments of V inherit closed-form dependence on the moments of X .

Steady-state moments. For wide-sense-stationary input with mean m , variance s^2 , autocovariance $\gamma(j)$,

$$\mathbb{E}[V] = \frac{Wm}{1-\lambda} = \tau Wm, \quad \text{Var}[V] = \frac{(Ws)^2}{1-\lambda^2} + 2W^2 \sum_{j \geq 1} \gamma(j) \frac{\lambda^j}{1-\lambda^2}. \quad (3)$$

The mean tracks the input rate amplified by τ ; the variance tracks the input variance plus an autocovariance term that depends on the *time ordering*, not the marginal. These are the levers each corruption pulls (made rigorous as Corollaries 1–4, with an explicit bound on the reset nonlinearity in Theorem 1; Appendix B).

Per-corruption signatures.

- **hot_pixel** adds a constant δ , shifting $\mathbb{E}[V]$ by $\tau W \delta$ until V saturates: affected μ_c leave the clean range (constant-drift instance of Corollary 1).
- **temporal_jitter** is a measure-preserving permutation: it leaves m, s^2 unchanged but perturbs $\gamma(j)$, converting the timing change into a s^2 shift that grows with layer depth (Corollary 3), so a deep, layer-4 view of the membrane is its strongest single detector (Section 4.7).
- **event_rate_shift** scales $m \rightarrow \alpha m$, displacing every channel’s mean along one global-activity direction, detectable by a 1-D projection once α leaves the clean band (the same mean-shift mechanism of (3), a multiplicative-drift instance of Corollary 1).
- **spatial_dropout** lowers σ_c^2 while μ_c stays intact, but GAP pools away the spatial dimension first, so the membrane looks *quieter and more typical* and one-sided distance scoring *inverts* (AUROC < 0.5) (the mirror-image, decreasing case of Corollary 2 with $\Sigma_\Delta \preceq 0$; the AUROC ceiling itself is Proposition 2).
- **event_flood** raises m nearly uniformly, the signature of a busy scene, giving near-chance per frame (the same mean-shift mechanism of Corollary 1); what distinguishes it lives in the spatial dispersion GAP removes and in sequence-level consistency (Appendix K).
- **polarity_flip** is the one exception: rather than following from the state equation, it rests on an empirical premise – that the trained detector is approximately polarity-symmetric – under which φ is near-invariant and the AUROC ceiling is ≈ 0.5 by construction (Proposition 2, Appendix B.3).

Taxonomy. The five signatures above group into three classes: *magnitude* (`hot_pixel`, `event_rate_shift`, `jitter-via-variance`), *structure* (`spatial_dropout`, the spatial part of `event_flood`), and *invariant* (`polarity_flip`, the one exception above, resting on network symmetry rather than the state equation). All five magnitude/structure predictions follow from (2)–(3) with no parameters fit to corruption labels.

3.4 SEVERITY MONOTONICITY AND THE SPIKE-MEMBRANE GAP

The qualitative predictions above sharpen into monotonicity guarantees: modelling a corruption as an additive current drift of severity γ and linearising the PLIF update in the sub-threshold regime (Appendix B), the membrane mean shifts at least linearly in γ (Corollary 1); any corruption that inflates the input covariance monotonically increases membrane variance (Corollary 2); and the expected Mahalanobis OOD score is non-decreasing in γ , strictly increasing whenever the feature mean shifts or its covariance grows (Corollary 4). This is the formal basis of the severity-monotonicity we measure (Spearman $\rho = +1$ for `hot_pixel`, `event_rate_shift`, `event_flood`; $\rho = +0.70$ for `temporal_jitter`, not significant at $n = 5$; Appendix N). The lone sign-flipped exception, `spatial_dropout` ($\rho = -1$), is exactly the structure-class inversion of the Section 3.3 taxonomy. The one nonlinearity this linearization sets aside – the impulsive reset – is bounded in proportion to the firing-disagreement rate (Theorem 1).

Finally, the membrane carries more information than the spikes it produces: below the firing threshold it strictly *contains* the spike counts (Example 1), and more generally the full trajectory V carries at least as much information as *any* spike statistic in every regime, since spikes are a deterministic, thresholded function of V (Corollary 5). The compact φ we read is a lossy summary of V , so this result ranks V , not φ , above spikes; under the stationary membrane model φ itself carries at least as much information as every *rate-based* spike statistic (spike-rate, spike-entropy; Appendix B), and empirically it also outperforms richer temporal spike statistics (Appendix G: φ mean AUROC 0.572 vs spike-rate 0.537).

3.5 THE MANIFOLD-DECOMPOSITION DETECTOR (MDD)

Every prior single detector (Mahalanobis, k -NN, GMM, OCSVM, PCA, MLP-AE, RealNVP) topped out near 0.57 macro (Figure 5) and inverted on contractions. MDD instead scores four orthogonal, two-sided axes, fused by a calibrated OR:

- **B1 radius** (two-sided isotropic energy): $||z|| - \mathbb{E}||z||/\text{sd}$ in a PCA- k space (shipped default $k=64$, insensitive over [16, 256], Appendix J).
- **B2 RCF** (Radial-Conditional): a *direction-conditioned* radius $||z|| - \mathbb{E}[||z|| \mid \text{dir}]/\text{sd}$ via cosine- k NN.
- **B3 L4 Mahalanobis** (one-sided) on the layer-4 $[\mu, \sigma^2, \kappa]$ block.
- **B4 spatial** (`phi_spatial`): Ledoit–Wolf Mahalanobis on the spatial-dispersion features.

Each branch is standardized on held-out clean and fused by a calibrated max. The shipped default folds an empirical-covariance Mahalanobis into B1 (`radius'`) and replaces the plain max with a studentized combiner $\max - \alpha \cdot \text{median}$ ($\alpha \approx 0.5$); the derivation and ablation are in Appendices H–I. An optional per-recording aggregation integrates the consistent per-sequence bias. Figure 2 shows the full pipeline.

4 EXPERIMENTAL RESULTS

4.1 SETUP

Prophesee Gen1 test split: 302 sequences, $\approx 343,099$ frames; each corruption applied per whole sequence at L1–L5. Detectors are fit on clean-train frames only and are corruption-blind; the split honours sequence boundaries so no clean frame appears in both fit and eval. We report AUROC (AUPR / FPR@95% at severity 5 in Table 4) per-frame and per- $W=64$ -window; all values are point estimates on a single split, with 95% bootstrap CIs on the headline numbers in Appendix M.

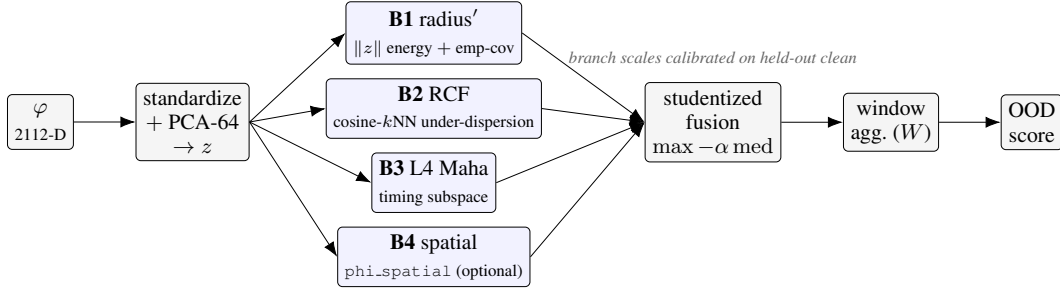


Figure 2: **The MDD pipeline** (described in Section 3.5): four two-sided branches over the PCA-denoised descriptor z , fused by a studentized $\max - \alpha$ median and optionally windowed.

4.2 MOTIVATION: A PRETRAINED DETECTOR DEGRADES UNDER EVENT CORRUPTIONS

We feed the pretrained hybrid SNN-ANN detector (Ahmed et al., 2025) (clean Gen1 test mAP ≈ 0.36 , Table 1) each Gen1-C corruption with no retraining, corrupting the event histogram before it is fed into the model. `event_flood` is the most destructive (mAP $0.341 \rightarrow 0.003$); `polarity_flip`

Corruption	S1	S2	S3	S4	S5
hot_pixel	0.348	0.324	0.267	0.216	0.159
event_flood	0.341	0.288	0.140	0.017	0.003
temporal_jitter	0.353	0.348	0.344	0.333	0.302
event_rate_shift	0.359	0.358	0.233	0.233	0.168
polarity_flip	0.357	0.356	0.352	0.348	0.344
spatial_dropout	0.357	0.351	0.344	0.323	0.265

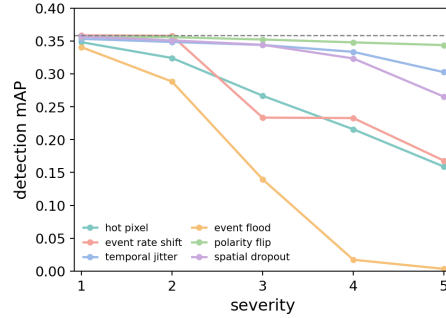


Table 1: Detection mAP (COCO AP@[.5:.95]) of the pretrained hybrid detector under each Gen1-C corruption at severities 1–5 (no retraining; clean baseline 0.358), and the same data plotted (right). `event_flood` degrades fastest; `polarity_flip` stays flat despite being undetectable by the membrane.

is nearly harmless (-4% at S5) and sits at AUROC ≈ 0.43 (Section 5).

4.3 MAIN RESULT: MDD OOD DETECTION

Table 2 reports fused MDD AUROC per corruption/severity, per-frame (a) and per-64-window (b, $W=64$, ≈ 0.64 s). One unsupervised, corruption-blind score is already strong per frame on three of the six corruptions at severity 5; aggregating over the bounded 64-frame window (Section 4.8) brings a fourth, `event_flood`, past 0.85 without whole-recording pooling. The two residuals are theory-predicted (Section 5).

4.4 EFFICIENT MDD

The result above still scores against the full 204,096-frame clean reference with a hard top-64 cosine search. To make MDD deployable on-device, we shrink the reference to 200 points chosen by farthest-point sampling (far better than random subsampling at matched size) and replace the hard top- k search with a TopK-free softmax-weighted average, since embedded ONNX toolchains commonly reject TopK. This costs only 0.010 mean AUROC ($0.802 \rightarrow 0.792$). We exported the resulting model with static shapes and measured real (not simulated) latency on phone-class NPUs/CPU's and two Arm Cortex-M7 microcontrollers: phone-class devices clear the 50 ms/frame real-time deadline by a wide margin (under 1 ms), and one of the two MCU boards does too (34.93 ms on the STM32H7S78-DK), but the other (STM32H735G-DK, a slightly slower 550 MHz core) comes

(a) per-frame						(b) per- $W=64$ -window (≈ 0.64 s)					
Corruption	S1	S2	S3	S4	S5	Corruption	S1	S2	S3	S4	S5
hot_pixel	1.000	1.000	1.000	1.000	1.000	hot_pixel	0.999	0.999	0.999	1.000	1.000
temporal_jitter	0.658	0.700	0.742	0.799	0.850	temporal_jitter	0.679	0.731	0.759	0.816	0.869
event_rate_shift	0.510	0.537	0.701	0.716	0.855	event_rate_shift	0.514	0.556	0.732	0.746	0.896
spatial_dropout	0.500	0.501	0.502	0.517	0.560	event_flood	0.556	0.612	0.690	0.790	0.889
event_flood	0.504	0.509	0.517	0.531	0.551	spatial_dropout	0.500	0.513	0.502	0.521	0.576
polarity_flip	0.504	0.508	0.516	0.527	0.540	polarity_flip	0.514	0.537	0.543	0.563	0.579

Table 2: Fused MDD AUROC (shipped improved default: $\text{radius}' + \text{studentized fusion}$). (a) per-frame: contractions and the invariant `polarity_flip` sit near chance. (b) per- $W=64$ -window: four corruptions reach ≥ 0.85 without full-recording pooling; `spatial_dropout` and `polarity_flip` remain the residuals.

Table 3: Measured (not simulated) inference latency of the embedded MDD variant on real hardware, against the 50 ms/frame deadline. Phone-class devices measured via Qualcomm AI Hub (Qualcomm Technologies, Inc., 2024); the STM32 board via ST Edge AI Developer Cloud’s (STMicroelectronics, 2024) hardware farm.

Device	Compute unit	Class	Measured latency
Samsung Galaxy S24	NPU	phone SoC	0.137 ms
Samsung Galaxy S24	CPU	phone SoC	0.101 ms
Samsung Galaxy A14 5G	CPU	phone SoC (budget)	0.823 ms
Google Pixel 3a	CPU	phone SoC (oldest tested)	1.091 ms
STM32H7S78-DK	CPU	MCU, Arm Cortex-M7 @ 600 MHz	34.93 ms
STM32H735G-DK	CPU	MCU, Arm Cortex-M7 @ 550 MHz	52.42 ms [†]

[†] The one board tested that *misses* the 50 ms/frame deadline ($1.048\times$ over budget).

in at 52.42 ms and *misses* the deadline (Table 3) — board-level differences, not just clock speed, decide whether a bare Cortex-M7 clears real-time on this workload. The exported graph takes both φ (2112-D) and `phi_spatial` (1408-D) per frame and, per ST Edge AI Core’s static analysis, costs 2.65M MACCs, 30.06 KiB of RAM, and 10.03 MiB of flash — 15–18 $\times$ under either board’s available internal RAM, with the flash placed in the external flash both boards carry. We did not separately profile a `phi_spatial`-free variant on hardware, so the reported cost is for the full (spatial-branch included) model; isolating the spatial branch’s incremental on-device delta is future work. Full derivation and the reference-reduction ablation are in Appendix C.

4.5 THE GEOMETRY OF CORRUPTION

Figure 11 shows the geometry underlying Section 3.3. Membrane energy $\|z\|$ moves *right* for dilations and *left* for contractions (Figure 11a); the Mahalanobis d^2 of the contractions sits at *smaller* values than clean (Figure 11b), the AUROC < 0.5 inversion; and the clean $\rightarrow$ corrupt centroid drift in a shared clean-PCA separates the corruption directions, with the two dilations (`hot_pixel`, `event_flood`; cosine 0.75) and the two contraction-side residuals (`polarity_flip`, `spatial_dropout`; 0.73) each sharing a direction. Figure 11 (radial energy and Mahalanobis d^2) and the PCA drift-arrow/cosine views are both in Appendix O.

4.6 REPRESENTATION ABLATION: WHICH SIGNAL FROM THE FORWARD PASS?

The geometry above operates on full membrane $\varphi = [\mu, \sigma^2, \kappa]$. We also check whether a cheaper slice of φ , or a different signal from the *same* forward pass, could substitute: no single moment or alternative read-out (spike-rate, spike-entropy, ANN features, cls-head features) matches full membrane, each dropping a specific corruption’s signal (e.g. spike-rate is blind to sub-threshold shifts, Example 1). Full membrane wins overall (mean AUROC 0.572); full per-corruption breakdown and methodology in Appendix G (Table 10).

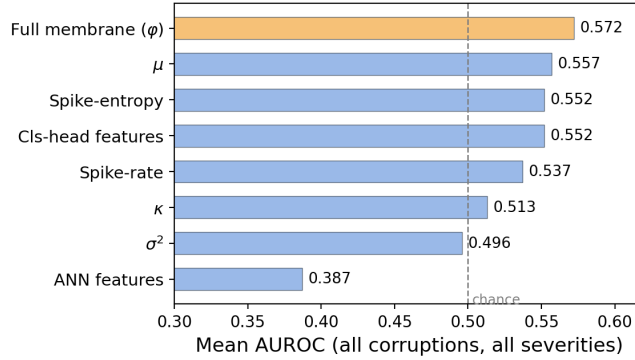


Figure 3: **Representation ablation**, mean AUROC over all corruptions and severities. Full membrane ϕ leads; ANN features are the weakest read-out from the same forward pass. Per-corruption breakdown in Appendix G.

4.7 BRANCH DECOMPOSITION AND FUSION ABLATION

Figure 4a shows each branch is corruption-specialized (radius owns dilations, L4 owns timing, RCF/spatial own contractions); the plain calibrated `max` sits *below* the best single branch on both residuals, motivating the studentized fusion (Appendix L). This follows the classical outlier-ensemble bias-variance argument (Aggarwal & Sathe, 2015): when an anomaly is flagged by only a *few* base detectors (here, one branch per corruption; leave-one-out, Table 12), averaging or consensus combiners are the wrong regime — they dilute the one branch that fired — and a variance-corrected `max` is the theoretically preferred aggregator; our 16-combiner benchmark (Appendix I) confirms this empirically, with every averaging/consensus/rank-aggregation/copula combiner underperforming the `max` family. Folding an empirical-covariance Mahalanobis into the radius branch and replacing `max` with `max` $-\alpha \cdot \text{median}$ recovers `spatial_dropout` from its plain-`max` baseline to 0.620 per sequence (0.576 at $W=64$) and lifts the worst-corruption floor with no easy-corruption regression, though still short of the `rcf` branch’s own 0.693 ceiling (Table 11); pushing α further approaches that ceiling (0.689) only by trading away `polarity_flip` (Appendix L, Figure 7a).

4.8 WINDOWED AGGREGATION: THE DEPLOYABLE OPERATING POINT

Table 2 (main text) is the headline: $W=64$ (≈ 0.64 s per decision) recovers most of the signal *without* any whole-recording assumption; full-sequence aggregation (~ 730 frames, once per recording) is an optimistic upper bound (Appendix M, Table 14). The W sweep at severity 5 shows when the signal becomes reliable: `hot_pixel` is at 1.00 by $W=16$ (≈ 0.16 s), all four detectable corruptions exceed 0.85 by $W=64$, and `event_flood` reaches 0.98 by $W=128$.

Windowing versus per-frame signal. `event_flood`’s rise ($0.55 \rightarrow 0.98$) comes from integrating a temporally-consistent bias, so we report the *windowed* value. `polarity_flip` and `spatial_dropout` stay flat at every window: pooling cannot recover a per-frame signal that is not there.

5 LIMITATIONS AND THE OPEN CHALLENGE

Residuals. Two corruptions remain near chance when detected from the membrane alone, but for different reasons, and across these six corruptions how detectable a corruption is tracks how much harm it does. `spatial_dropout` causes real harm (-26% mAP at S5, Section 4.2) but leaves no per-frame membrane signature, because that signature is *spatial* and global average pooling throws spatial layout away. The spatial-dispersion features we release (`phi_spatial`) feed MDD’s spatial branch, and we validated end-to-end that they do recover this signal on a controlled test case (a signal present only in `phi_spatial` lifts AUROC from ≈ 0.50 , without the spatial branch, to ≈ 1.00 ; Appendix K). So this is a genuine **open problem**: a harmful corruption with no cheap per-frame

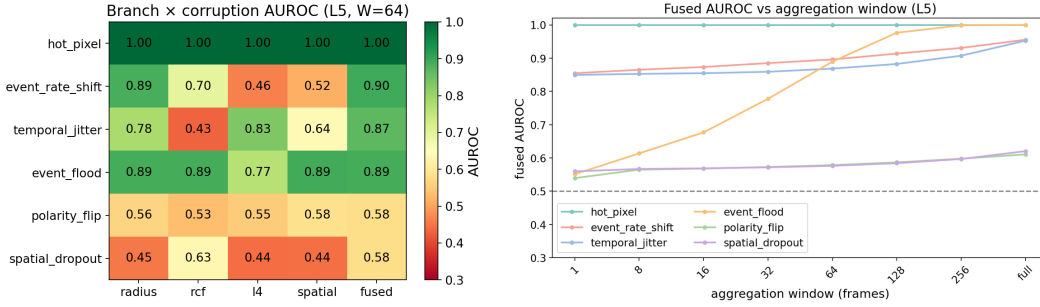


Figure 4: MDD detection at L5. (a) per-branch AUROC (corruption × branch, $W=64$): each branch owns a different corruption and fused tracks the per-corruption best. (b) fused AUROC vs. aggregation window W : the four detectable corruptions rise with W (event_flood slowest, crossing 0.85 near $W=128$) while the two residuals stay flat at every window. The α trade-off is in Appendix L (Figure 7a); the combiner benchmark is in Appendix I.

signal, and the gap is capped by a provable ceiling on any φ -based detector, not just a tuning failure: no score computed from φ can exceed $\text{AUROC} \leq \frac{1}{2} + \delta$, where δ is the total variation distance between the clean and corrupted φ distributions (Proposition 2). polarity_flip is a different case: a property of the trained network’s polarity invariance, not of our representation. It causes negligible harm (-4% at S5) and cannot be detected from V_{mem} by construction, since the model itself is unaffected: the same network invariance that makes a corruption harmless also makes it invisible to φ , separating polarity_flip (an invariance) from spatial_dropout (harmful but hidden by the pooling blind spot instead); see Proposition 2.

Other limitations. (i) The low-compute claim is only about the *signal*, not the MDD detector that scores it (host-side cost profiled in §4.4, Appendix C); notably, the reference-set reduction there lands almost entirely on spatial_dropout (-0.106 , Table 6) – shrinking MDD for deployment makes the open problem above worse, not better. (ii) φ is read from the membrane state including the reset that follows each spike, so the linear, sub-threshold theory in Section 3.3 is only an approximation; a version of V with spikes strictly excluded, and a reset-aware analysis, remain untested. (iii) Our AUROC values are point estimates from a single data split; we report 95% bootstrap confidence intervals (resampling recordings) for the headline numbers in Appendix M. (iv) We use a single dataset (Gen1) and a single pretrained host model; transfer to other datasets (DSEC, Gen4) or architectures is untested. The corruption severities are synthetic, monotone transforms chosen to resemble real sensor faults (Section 3.2, Appendix D), but they are *not* calibrated against measured hardware faults; a realistic-noise simulator such as v2e (Hu et al., 2021), or capturing faults directly from hardware, would validate this mapping. What we contribute here is the benchmark, protocol, and feature release. (v) The headline numbers use a 64-frame window (Table 2); the full-recording upper bound (Table 14) should not be read as a per-frame detection rate. AUROC measures how separable clean and corrupted scores are once the ground-truth labels are known, not a value the detector would reproduce at a fixed deployment threshold, where no such labels are available.

6 CONCLUSION

Gen1-C, the leaky-integrator theory, and MDD together turn a signal that costs nothing extra to read into a detector whose successes and residuals are both predicted in advance rather than discovered post hoc. The one open problem, spatial_dropout, points to spatial-dispersion features and adaptive neurons as the natural next directions.

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APPENDIX

This appendix documents the complete experimental program – including the levers that did *not* work – so the headline claims can be audited against everything we tried.

A IMPLEMENTATION, SPLITS, AND REPRODUCIBILITY

Feature extraction. A forward hook on all four parametric LIF layers (64/128/256/256, 704 channels) captures $V \in \mathbb{R}^{T \times 1 \times C \times H \times W}$, GAPs over space, and reduces to $[\mu, \sigma^2, \kappa]$ per channel (φ , 2112-D). `phi_spatial` (1408-D) adds per-channel spatial variance and participation ratio, stored `float32` (`float16` overflows the spatial-variance channel under `event_flood`).

Protocol. Metrics AUROC, AUPR, FPR@95%, per-frame and per-sequence, with 95% bootstrap CIs on the headline numbers (Appendix M).

Corruption (S5)	per-frame		per- $W=64$ -window	
	AUPR	FPR@95%	AUPR	FPR@95%
hot_pixel	1.000	0.000	1.000	0.000
temporal_jitter	0.867	0.657	0.893	0.678
event_rate_shift	0.862	0.685	0.913	0.641
event_flood	0.603	0.944	0.928	0.767
spatial_dropout	0.551	0.944	0.571	0.940
polarity_flip	0.548	0.936	0.589	0.933

Table 4: Fused MDD at severity 5: AUPR and FPR@95%, per-frame and per- $W=64$ -window, computed from the same fitted branches and studentized fusion as Table 2 (whose S5 AUROC column applies unchanged here).

Reproduction budget. The one-time GPU cost is extraction (~ 90 – 150 GB of φ); every downstream stage then runs on φ directly in minutes to an hour on one GPU.

B MONOTONICITY THEORY: STATEMENTS AND PROOFS

We make the qualitative predictions of Section 3.3 *quantitative*. Let severity be $\gamma \geq 0$ and model a corruption family as an additive drift on the per-layer input current:

$$X_t(\gamma) = X_t(0) + \gamma \Delta_t + \epsilon_t(\gamma), \quad \mathbb{E}[\epsilon_t] = 0. \quad (4)$$

Writing the membrane deviation $\delta V_t(\gamma) = V_t(\gamma) - V_t(0)$ and linearising the PLIF update $V[t] = (1 - \frac{1}{\tau})V[t-1] + WX[t]$ in the sub-threshold regime (bounded reset mismatch) gives

$$\delta V_t(\gamma) = \lambda \delta V_{t-1}(\gamma) + W(\gamma \Delta_t + \epsilon_t), \quad (5)$$

which unrolls to

$$\delta V_t(\gamma) = \sum_{k=0}^t \lambda^{t-k} W(\gamma \Delta_k + \epsilon_k). \quad (6)$$

Everything below is one of two kinds of result: how this linear recursion propagates a monotone perturbation (Lemma 1, Corollaries 1–4), or how much information the membrane retains relative to the spikes it produces (Lemma 2, Corollaries 5–6 and Proposition 1). The reset nonlinearity both families treat as a bounded perturbation is quantified first.

Reset-aware dynamics and the error term R_t . The linearisation (5) is *exact between spikes*; the sole nonlinearity is the hard reset, which subtracts the threshold θ from the trajectory at each emission. With the spike indicator $s_t = 1[V_t \geq \theta]$ the reset-aware recurrence is

$$V[t] = \left(1 - \frac{1}{\tau}\right)V[t-1] + WX[t] - \theta s_{t-1}, \quad (7)$$

so the membrane deviation δV_t of (6) acquires exactly one extra, *impulsive* term, the difference of the two reset trains:

$$R_t = -\theta \sum_{k=0}^t \lambda^{t-k} (s_k^{\text{OOD}} - s_k^{\text{ID}}), \quad \lambda := 1 - \frac{1}{\tau}. \quad (8)$$

This is the PLIF nonlinearity written out, not a small-signal approximation: the dynamics are a *piecewise-linear hybrid system*, linear on inter-spike intervals plus a bounded jump at each reset. The jump term is controlled by the firing statistics.

Theorem 1 (Reset error scales with the firing rate). *Let $\rho_\Delta = \sup_k \Pr[s_k^{\text{OOD}} \neq s_k^{\text{ID}}]$ be the per-step probability that the clean and corrupted spike trains disagree. Then*

$$\mathbb{E}|R_t| \leq \frac{\theta \rho_\Delta}{1 - \lambda}. \quad (9)$$

In particular $\rho_\Delta = 0$ in the sub-threshold regime, recovering $R_t \equiv 0$ and the exact linear results below, and the membrane-mean signal b of Corollary 1 dominates the reset error whenever $|b| \gg \theta \rho_\Delta / (1 - \lambda)$.

Proof. By the triangle inequality $|R_t| \leq \theta \sum_{k=0}^t \lambda^{t-k} |s_k^{\text{OOD}} - s_k^{\text{ID}}|$. Taking expectations and using $\mathbb{E}|s_k^{\text{OOD}} - s_k^{\text{ID}}| = \Pr[s_k^{\text{OOD}} \neq s_k^{\text{ID}}] \leq \rho_\Delta$ and the geometric sum $\sum_{k \leq t} \lambda^{t-k} \leq (1 - \lambda)^{-1}$ gives (9). $\square$

Theorem 1 makes the “bounded reset mismatch” assumption of Corollaries 1–4 *quantitative*: the constant C_R is $\theta\rho_\Delta/(1 - \lambda)$, the guarantees are exact in the low-firing regime where the membrane signal is most distinctive, and they degrade gracefully, linearly in the firing-disagreement rate ρ_Δ , as neurons fire and reset more often.

B.1 LINEAR-RESPONSE MONOTONICITY

The next four results are all instances of one fact: a stable linear recursion (or, one level up, a quadratic form of a linearly-drifting feature vector) turns a monotone perturbation of its input into a monotone perturbation of its output. Stating that fact once, as a lemma, lets each corollary below be a two-line specialization rather than a repeated derivation.

Lemma 1 (Linear-response monotonicity). *(a) Membrane moments.* Let $\delta V_t(\gamma)$ solve the linearised PLIF recursion (6) for a per-channel current whose mean or covariance is affine in γ : either $\mathbb{E}[\Delta_k] \neq 0$ under (4) (mean-drift), or

$$\text{Cov}(X_t(\gamma)) = \Sigma_0 + \gamma \Sigma_\Delta, \quad \Sigma_\Delta \succeq 0 \quad (10)$$

(covariance-inflation). Then, up to the bounded reset error R_t of Theorem 1,

$$\mathbb{E}[\delta V_t(\gamma)] = \gamma \sum_{k=0}^t \lambda^{t-k} W \mathbb{E}[\Delta_k] + R_t, \quad \text{monotone in } \gamma \text{ whenever the sum} \neq 0, \quad (11)$$

$$\Sigma_V(\gamma) = \frac{W^2}{1 - \lambda^2} (\Sigma_0 + \gamma \Sigma_\Delta), \quad \frac{d}{d\gamma} \text{tr} \Sigma_V(\gamma) = \frac{W^2 \text{tr}(\Sigma_\Delta)}{1 - \lambda^2} \geq 0. \quad (12)$$

(b) Any linearly-drifting quadratic form. Let z_γ be a feature vector with

$$\mathbb{E}[z_\gamma] = m_0 + \gamma b, \quad \text{Cov}(z_\gamma) = C_0 + \gamma C_\Delta, \quad C_\Delta \succeq 0. \quad (13)$$

Then for the quadratic form $D^2(z_\gamma) := (z_\gamma - m_0)^\top C_0^{-1} (z_\gamma - m_0)$,

$$\mathbb{E}[D^2(z_\gamma)] = n + \gamma \text{tr}(C_0^{-1} C_\Delta) + \gamma^2 b^\top C_0^{-1} b, \quad (14)$$

non-decreasing in $\gamma \geq 0$ and strictly increasing whenever $b \neq 0$ or $C_\Delta \neq 0$.

Proof. (a) Taking expectations in (6) with $\mathbb{E}[\epsilon_k] = 0$ gives the first line of (11) directly, with R_t collecting the reset-mismatch error ($|R_t| \leq C_R$ in the sub-threshold regime). For (12), model each channel as a stable AR(1) process $V[t] = \lambda V[t-1] + WX[t]$; at stationarity the variance solves the discrete Lyapunov equation $\Sigma_V = \lambda^2 \Sigma_V + W^2 \Sigma_X$, i.e. $\Sigma_V = W^2 \Sigma_X / (1 - \lambda^2)$. Substituting $\Sigma_X = \Sigma_0 + \gamma \Sigma_\Delta$ and differentiating the trace gives $\text{tr}(\Sigma_\Delta) \geq 0 \Rightarrow d/d\gamma \text{tr} \Sigma_V(\gamma) \geq 0$. (b) Write $z_\gamma = m_0 + \gamma b + \xi_\gamma$ with $\mathbb{E}[\xi_\gamma] = 0$, $\text{Cov}(\xi_\gamma) = C_0 + \gamma C_\Delta$. Expanding the quadratic form,

$$\begin{aligned} \mathbb{E}[D^2(z_\gamma)] &= \mathbb{E}[(\gamma b + \xi_\gamma)^\top C_0^{-1} (\gamma b + \xi_\gamma)] \\ &= \gamma^2 b^\top C_0^{-1} b + \underbrace{2\gamma b^\top C_0^{-1} \mathbb{E}[\xi_\gamma]}_{=0} + \mathbb{E}[\xi_\gamma^\top C_0^{-1} \xi_\gamma] \\ &= \gamma^2 b^\top C_0^{-1} b + \text{tr}(C_0^{-1} (C_0 + \gamma C_\Delta)) = n + \gamma \text{tr}(C_0^{-1} C_\Delta) + \gamma^2 b^\top C_0^{-1} b, \end{aligned}$$

using the trace identity $\mathbb{E}[\xi^\top A \xi] = \text{tr}(A \text{Cov}(\xi))$ for zero-mean ξ . Both coefficients are non-negative ($C_0^{-1} \succ 0$, $C_\Delta \succeq 0$, so $\text{tr}(C_0^{-1} C_\Delta) \geq 0$ and $b^\top C_0^{-1} b \geq 0$), giving (14). $\square$

Corollary 1 (Monotone membrane drift). *Under (4) with $\mathbb{E}[\Delta_k] \neq 0$ in at least one monitored PLIF layer and bounded reset mismatch, Lemma 1(a)/(11) gives*

$$\mathbb{E}[\delta V_t(\gamma)] = \gamma b + R_t, \quad b := \sum_{k=0}^t \lambda^{t-k} W \mathbb{E}[\Delta_k], \quad |R_t| \leq C_R. \quad (15)$$

The temporal-mean coordinate μ of φ is therefore OOD-informative whenever $b \neq 0$.

Corollary 2 (Monotone membrane variance). *Under the covariance-inflation hypothesis (10), Lemma 1(a)/(12) gives a stationary membrane covariance $\Sigma_V(\gamma)$ whose trace is non-decreasing in γ . Any corruption that increases the input covariance in the PSD order therefore monotonically increases the membrane variance, justifying the σ^2 and κ coordinates of φ beyond the mean.*

Corollary 3 (Marginal-preserving reordering shifts membrane variance). *Let a corruption T_γ permute event timing within a bounded window, leaving the per-bin marginal mean and variance m, s^2 unchanged but perturbing the input autocovariance $\gamma(j) \rightarrow \gamma(j) + \Delta\gamma(j)$ for $j \geq 1$. Channeling the perturbation of Lemma 1(a) through the autocovariance term of the steady-state variance (3) rather than Σ_0 directly, the membrane variance shifts by*

$$\Delta \text{Var}[V] = \frac{2W^2}{1-\lambda^2} \sum_{j \geq 1} \Delta\gamma(j) \lambda^j, \quad (16)$$

whose magnitude is monotone increasing in $\lambda = 1 - 1/\tau$ for every fixed j : deeper layers (larger τ) amplify the same autocovariance perturbation into a larger σ^2 shift, even though the input’s marginal moments are unchanged.

Proof. Since m, s^2 are held fixed by construction, only the autocovariance term changes under T_γ , giving (16) directly by linearity. Differentiating the j -th summand’s coefficient with respect to λ ,

$$\frac{d}{d\lambda} \frac{\lambda^j}{1-\lambda^2} = \frac{\lambda^{j-1} [j + (2-j)\lambda^2]}{(1-\lambda^2)^2},$$

and the bracket is affine in λ^2 with value $j > 0$ at $\lambda^2 = 0$ and value $2 > 0$ at $\lambda^2 = 1$, hence strictly positive throughout $\lambda \in (0, 1)$ for every integer $j \geq 1$: each term of (16) is strictly increasing in λ . $\square$

Corollary 4 (MP-OOD score monotonicity). *Let the clean calibration model be (m_0, C_0) and suppose φ_γ satisfies the $z_\gamma = \varphi_\gamma$ instance of (13):*

$$\mathbb{E}[\varphi_\gamma] = m_0 + \gamma b, \quad \text{Cov}(\varphi_\gamma) = C_0 + \gamma C_\Delta, \quad C_\Delta \succeq 0. \quad (17)$$

By Lemma 1(b)/(14), the expected Mahalanobis OOD score $D^2(\varphi_\gamma) = (\varphi_\gamma - m_0)^\top C_0^{-1} (\varphi_\gamma - m_0)$ satisfies

$$\mathbb{E}[D^2(\varphi_\gamma)] = n + \gamma \text{tr}(C_0^{-1} C_\Delta) + \gamma^2 b^\top C_0^{-1} b, \quad (18)$$

non-decreasing in $\gamma \geq 0$, strictly increasing whenever $b \neq 0$ or $C_\Delta \neq 0$. The lone measured exception, `spatial_dropout` ($\rho = -1$), is the failure of (17) when GAP pools away the dimension in which b lives: the score then sees only a contraction toward m_0 and the sign flips.

B.2 DATA-PROCESSING DOMINANCE OF THE MEMBRANE OVER SPIKE STATISTICS

The next results are all instances of a second shared fact: if a statistic is a deterministic function of a richer object, the data-processing inequality bounds how much it can know about any external parameter relative to that richer object.

Lemma 2 (Data-processing dominance). *Let $Y = h(Z)$ for a deterministic, measurable map h , and let γ be any parameter (or label) with a prior. Then for any post-hoc statistic g ,*

$$I(\gamma; g(Y)) \leq I(\gamma; Y) \leq I(\gamma; Z), \quad (19)$$

with equality on the right iff $\gamma \perp Z \mid Y$.

Proof. $Y = h(Z)$ makes $\gamma \rightarrow Z \rightarrow Y$ Markov, so the data-processing inequality gives $I(\gamma; Y) \leq I(\gamma; Z)$; applying it again to the further map g gives the left inequality. Equality holds iff $\gamma \rightarrow Y \rightarrow Z$ is also Markov, i.e. $Z \perp \gamma \mid Y$. $\square$

Corollary 5 (Membrane information dominates spikes in every regime). *The spike train is a deterministic function of the membrane trajectory, $s_{1:T} = h(V_{1:T})$ with $h(v)_t = \mathbf{1}[v_t \geq \theta]$. Instantiating Lemma 2 with $Z = V_{1:T}$, $Y = s_{1:T}$, for any post-hoc spike statistic g ,*

$$I(\gamma; g(s_{1:T})) \leq I(\gamma; s_{1:T}) \leq I(\gamma; V_{1:T}). \quad (20)$$

A membrane-based detector therefore has at least as much information about the corruption as any spike-count detector, with strict gain whenever the corruption perturbs the membrane other than through threshold crossings.

Example 1 (Sub-threshold shifts are invisible to spike counts). Concretely: a neuron with threshold θ , in-distribution potential in $[0, \theta - \varepsilon]$, and a corruption shifting every membrane value by $0 < \delta < \varepsilon$. Then

$$V_t^{\text{OOD}} = V_t^{\text{ID}} + \delta \leq \theta - \varepsilon + \delta < \theta, \quad (21)$$

so neither trajectory fires ($s_t^{\text{ID}} = s_t^{\text{OOD}} = 0$ for all t) and every spike-count detector assigns identical scores to the two distributions. Yet by linearity of expectation $\bar{V}^{\text{OOD}} - \bar{V}^{\text{ID}} = \delta$, so a membrane-potential detector (e.g., Mahalanobis on φ) separates them directly: sub-threshold membrane OOD strictly subsumes spike-count OOD. This is the equality case of Corollary 5 being tight ($I(\gamma; s_{1:T}) = 0 < I(\gamma; V_{1:T})$), not a separate argument.

This holds under resets, nonlinearities, and arbitrary firing rates: spiking is a lossy coarsening of the membrane. One scope caveat: the bound orders the *full* trajectory $V_{1:T}$; the deployed φ is itself a lossy map of V , so Lemma 2 does not by itself order φ against spikes – the next result closes that gap under a stationarity hypothesis. φ -dominance is provable for the sub-threshold-shift family (Example 1) and, under the stationary model, for every *rate-based* spike statistic (Corollary 6); only against statistics exploiting temporal structure beyond the marginal is it *empirical* (Appendix G: φ mean AUROC 0.572 vs spike-rate 0.537). Corollaries 1–4 additionally assume stationarity and the bounded reset mismatch of Theorem 1.

Corollary 6 (φ dominates rate-based spike statistics under location–scale). *Adopt the wide-sense-stationary membrane model of Corollaries 1–2 and suppose each PLIF unit’s sub-threshold potential is location–scale: $V_t \stackrel{d}{=} \mu + \sigma Z$ for a fixed standardized law $Z \sim F_0$ (mean 0, variance 1), with (μ, σ^2) the coordinates φ records. Then the mean firing rate is a deterministic function of φ ,*

$$r = \Pr[V_t \geq \theta] = \bar{F}_0\left(\frac{\theta - \mu}{\sigma}\right) = h(\varphi), \quad \bar{F}_0 := 1 - F_0, \quad (22)$$

so instantiating Lemma 2 with $Z = \varphi$, $Y = r = h(\varphi)$, for any post-hoc rate statistic g – in particular the spike-rate r itself and the spike-entropy $H(r) = -r \log r - (1 - r) \log(1 - r)$ –

$$I(\gamma; g(r)) \leq I(\gamma; r) \leq I(\gamma; \varphi). \quad (23)$$

The right inequality is strict whenever γ moves (μ, σ) off a level set of $(\theta - \mu)/\sigma$ (e.g. the variance/kurtosis inflation of temporal-jitter at $\mu \approx \text{const}$), which leaves every rate statistic fixed while moving φ . The Gaussian marginal of Corollary 2 is the special case $F_0 = \Phi$; the hypothesis needed is only that a fixed tail shape is rescaled by (μ, σ) , strictly weaker than Gaussianity and directly checkable on the sub-threshold histograms.

Proof. Stationarity gives V_t a fixed marginal law; the location-scale hypothesis writes it as $\mu + \sigma Z$, so $\Pr[V_t \geq \theta] = \Pr[Z \geq (\theta - \mu)/\sigma] = F_0((\theta - \mu)/\sigma)$, a measurable function of φ 's coordinates, giving (22). Hence $r = h(\varphi)$ is conditionally degenerate given φ , and Lemma 2 yields (23). For strictness, take a γ -perturbation whose induced $(\hat{\mu}, \hat{\sigma})$ lies in the kernel of $\nabla[(\theta - \mu)/\sigma]$ but not in the kernel of the φ -map: r (hence any $g(r)$) is unchanged while φ moves, so $I(\gamma; r) < I(\gamma; \varphi)$. $\square$

This explains *why* the rate baselines are weak: a rate is a single monotone function of $(\mu - \theta)/\sigma$, confounding the mean and variance shifts that φ resolves separately. (Statistics reading temporal correlation beyond the marginal are ordered by V , not necessarily φ , per Corollary 5.)

Dropping the location-scale hypothesis costs the clean identity $r = h(\varphi)$ (two marginals with the same (μ, σ^2) can put different mass past θ), but buys a *reason* for the kurtosis coordinate, which otherwise sits in φ without theoretical justification – and it is no longer a corollary of Lemma 2, since r is no longer a deterministic function of φ :

Proposition 1 (Moment-pinned firing rate: why κ enters φ). *Fix only the moments φ records. For a stationary marginal with mean μ , variance σ^2 , and $\theta > \mu$, the firing rate $r = \Pr[V_t \geq \theta]$ lies in a moment-determined interval $r \in [\underline{r}(\varphi), \bar{r}(\varphi)]$ rather than being pinned exactly. With two moments, the one-sided Cantelli bound gives*

$$r \leq \frac{\sigma^2}{\sigma^2 + (\theta - \mu)^2}, \quad (24)$$

and adjoining the fourth moment (the κ coordinate) strictly narrows the interval via the classical four-moment Chebyshev–Markov system: enlarging the fixed-moment set can only shrink the feasible family of marginals, hence the extremal range $\Delta(\varphi) = \bar{r} - \underline{r}$. Thus r is φ -measurable up to a slack $\Delta(\varphi)$ that κ reduces.

Proof. Eq. (24) is Cantelli’s inequality applied to $V - \mu$ at level $\theta - \mu > 0$. The interval $[\underline{r}, \bar{r}]$ is the range of $\Pr[V \geq \theta]$ over all laws matching the fixed moments – a Chebyshev–Markov moment problem whose extrema are attained by finitely-supported laws; the consistent fourth-moment constraint restricts that feasible set, so the interval is non-increasing in the number of fixed moments. $\square$

This is the honest weakening of Corollary 6: it admits the rate is only *pinned*, not determined, by φ , and identifies κ as the coordinate that tightens the pinning. Converting the slack $\Delta(\varphi)$ into a quantitative information inequality $I(\gamma; r) \leq I(\gamma; \varphi) + \epsilon(\Delta)$ needs a continuity argument on the channel and is left as future work; the moment bound (24) itself is exact.

B.3 WHY THE DETECTOR’S BLIND SPOTS TRACK THE MODEL’S

The results above bound *detectability*. The next proposition ties detectability to *harm* and, in its third part, turns that link into a quantitative ceiling: a corruption the trained network is invariant to is, at the monitored layer, simultaneously harmless and φ -undetectable (part i); the two observed residuals fail for different reasons (part ii); and the harmful-but-silent case is bounded by a provable information ceiling, not just a tuning failure (part iii).

Proposition 2 (Invariance, harm, and the φ -information ceiling). *Let h be the monitored PLIF state of the trained network, $y = \Psi(h)$ its task output, and $\varphi = \Phi(h)$ the membrane read-out, both deterministic functions of h . For a corruption T_γ write $h_\gamma \stackrel{d}{=} h(T_\gamma X)$.*

- (i) **Invariance $\Rightarrow$ harmless and undetectable.** *If $h_\gamma \stackrel{d}{=} h_0$, then $I(\gamma; \varphi) = 0$ and every φ -detector is at chance: harmlessness and φ -undetectability are the same event at the monitored layer, not a coincidence.*
- (ii) **The two residuals are distinct failures.** *φ -detectability requires $h_\gamma \not\stackrel{d}{=} h_0$; harm requires the shift to lie in a direction Ψ is sensitive to. `polarity_flip` moves neither ($h_\gamma \approx h_0$: near-harmless and φ -silent by construction, an instance of part i). `spatial_dropout` is the opposite: h moves in a task-relevant direction (hence harm), but that direction lies in the spatial layout $\Phi = \text{GAP}$ discards, so φ is silent while h is not – a genuine gap, not an invariance.*
- (iii) **The gap in part ii is bounded, not open-ended.** *Let P, Q be the laws of per-frame φ under clean and `spatial_dropout` inputs, $\delta = \text{TV}(P, Q)$. Then every detector reading φ , of any form, fitted or not, obeys*

$$\text{Acc}^* \leq \frac{1}{2}(1 + \delta), \quad \text{AUROC}^* \leq \frac{1}{2} + \delta, \quad (25)$$

and δ is non-increasing under pooling: $\varphi = \text{GAP}(V) \Rightarrow \delta = \text{TV}(\text{GAP}_{\#P_V}, \text{GAP}_{\#Q_V}) \leq \text{TV}(P_V, Q_V)$, so the ceiling is set by what pooling keeps, not by the detector.

Proof. (i) $y = \Psi(h)$ and $\varphi = \Phi(h)$ are measurable functions of h , so $h_\gamma \stackrel{d}{=} h_0$ forces $\Psi(h_\gamma) \stackrel{d}{=} \Psi(h_0)$ and $\Phi(h_\gamma) \stackrel{d}{=} \Phi(h_0)$; equal laws carry zero mutual information with γ and give Bayes-optimal separation $\frac{1}{2}$ (chance AUROC). (ii) If φ separates γ then $\Phi(h_\gamma) \not\stackrel{d}{=} \Phi(h_0)$, hence $h_\gamma \not\stackrel{d}{=} h_0$. Harm depends on h only through Ψ , so a shift confined to $\ker \Psi$ is harmless and a shift confined to $\ker \Phi$ (e.g. GAP’s spatial null space) is φ -undetectable; the two kernels differ, so a corruption can inhabit one without the other. (iii) Le Cam’s two-point inequality: with equal priors the minimal average error of any test between P and Q is $\frac{1}{2}(1 - \delta)$, so accuracy is at most $\frac{1}{2}(1 + \delta)$. For the AUROC bound, any score $s(\varphi)$ pushes P, Q to laws μ, ν on $\mathbb{R}$ with $\text{TV}(\mu, \nu) \leq \delta$ (data-processing for total variation); writing the rank statistic (no ties) as $\text{AUROC} = \frac{1}{2}(1 + \mathbb{E} \text{sign}(V - U))$ for independent $U \sim \mu, V \sim \nu$, and using $\mathbb{E} \text{sign}(V - U) = \int (2F_\mu - 1) d(\nu - \mu) \leq \|2F_\mu - 1\|_\infty \|\nu - \mu\|_1 \leq 2\delta$, gives $\text{AUROC} \leq \frac{1}{2} + \delta$. The pooling statement is data-processing for TV applied to the deterministic map GAP. $\square$

This is the formal content of “detectability tracks harm”: the detector inherits the network’s invariances (part i); the lone harmful-yet-silent case is quarantined to a pooling null-space (part ii); and that gap is provably bounded, not just an unmet search (part iii). We do not estimate $\delta = \text{TV}(P, Q)$ directly; the per-frame φ -detectors already sit at $\text{AUROC} \approx 0.5$ on `spatial_dropout` (Table 2), consistent with a small- δ regime, so the failure is plausibly *information-limited* rather than detector-limited: a more cautious claim than “we could not find a detector”. Consistently, the un-pooled spatial-dispersion read-out (`phi_spatial`, which does *not* discard the layout δ lives in) lifts a controlled contraction from ≈ 0.50 to ≈ 1.00 (Appendix K): the signal is present in V and recoverable off the GAP null-space, exactly as Eq. (25) predicts.

C COMPUTE-COST COMPARISON

We separate two claims. The **signal** φ adds negligible *inference* compute: V is state the network *already computes*, so producing φ is an $O(\text{channels})$ read plus three moments, with no extra MACs, weight fetches, or second network. The **detector** (MDD) is a separate, non-zero cost: PCA, a Mahalanobis form, and a cosine- k NN against the clean reference, the same arithmetic the ANN-feature baselines (Lee et al., 2018; Sun et al., 2022) use. What separates the methods structurally

Table 5: Inference-time compute comparison. Every baseline adds non-trivial computation; Vmem- φ reads a byproduct of the existing forward pass and tests a compact per-frame vector (Ex. 1).

Method	Added computation	SNN obstacle
Vmem-φ (ours)	native V read + 3 moments; host-side test on 2112-D φ	<i>none</i> : V is a byproduct of the existing forward pass
Logit-based (Hendrycks & Gimpel, 2017; Liu et al., 2020; Liang et al., 2018; Sun et al., 2021; Wang et al., 2022a; Huang et al., 2021) (MSP, Energy, ODIN, ReAct, ViM, GradNorm)	softmax/log-sum-exp/gradients over logits, some requiring a backward pass	logits and penultimate float activations absent in spiking heads
ANN-feature (kNN, Mahalanobis) (Lee et al., 2018; Sun et al., 2022)	auxiliary ANN forward pass for features	second non-spiking feature extractor

Table 6: Fused AUROC ($W=64$, $L5$) as the RCF reference set and search are shrunk for embedded export. **Full**: all 204,096 clean frames, hard top-64 search. **FPS-200**: 200 farthest-point-sampled reference frames, hard top-64 search (still requires TopK). **FPS-200, softmax**: same 200 references, softmax-weighted average ($\tau=40$) replacing TopK — the variant actually exported and profiled below.

Corruption	Full (204k, top-64)	FPS-200 (top-64)	FPS-200 (softmax, $\tau=40$)	Δ (Full $\rightarrow$ embedded)
hot_pixel	1.000	1.000	1.000	0.000
event_flood	0.889	0.891	0.891	+0.002
temporal_jitter	0.869	0.918	0.891	+0.022
event_rate_shift	0.896	0.930	0.919	+0.023
polarity_flip	0.579	0.564	0.581	+0.002
spatial_dropout	0.576	0.483	0.470	-0.106
Mean	0.802	0.798	0.792	-0.010

is the *signal*, not the detector: φ is a byproduct of the forward pass the deployment already runs, whereas the ANN detectors first need readouts a spiking substrate does not natively emit (Table 5); on an object *detector* there is no class posterior for the logit methods at all. Measured, the full, unshrunk MDD runs at $\sim 2.7 \times 10^4$ frames/s (≈ 0.037 ms/frame) on a single NVIDIA RTX 4060 laptop GPU (extraction cost and the per-stage reproduction budget are in Appendix A; Table 5 gives the per-detector-family breakdown). This appendix also profiles a shrunk, embedded-deployable variant on real phone and microcontroller hardware below.

Shrinking MDD for embedded deployment. The GPU figure above still scores against the *full* 204,096-frame clean reference. We test whether that reference can be cut orders of magnitude without losing accuracy, and whether the resulting graph runs on real phone- and microcontroller-class silicon under the 50 ms/frame deadline set by the Gen1 histogram bin width ($\Delta t=50$ ms, stacked_histogram.dt=50_nbins=10).

Reference-set reduction. Random subsampling of the 204k reference costs accuracy as n_{ref} shrinks (Appendix J: mean AUROC $0.802 \rightarrow 0.783$ at $n_{\text{ref}}=1000$, driven almost entirely by spatial_dropout collapsing $0.576 \rightarrow 0.499$). Replacing random subsampling with farthest-point sampling (FPS) — greedily selecting the n_{ref} mutually most dissimilar reference directions by cosine distance — recovers full-reference accuracy at $1000\times$ fewer stored points: FPS-200 reaches mean AUROC 0.798, matching the full 204k reference (0.802) and beating even random-1000 (Table 6).

Removing the top- k search. FPS-200 still needs a top-64 cosine search per frame, an op embedded ONNX toolchains (ST Edge AI Core 4.0.1 (STMicroelectronics, 2024)) reject outright. Replacing it with a softmax-weighted average over the 200 reference points (temperature swept, best point $\tau=40$) yields a fully ONNX-exportable graph at a cost of 0.006 further mean AUROC ($0.798 \rightarrow 0.792$; Table 6).

Table 7: The six corruptions, the sensor failure each models, and the input axis it perturbs.

Corruption	Physical failure mode	Axis of the input perturbed
hot_pixel	defective / stuck pixels	constant DC offset at fixed pixel locations
event_flood	EMI / sensor saturation	localized count bursts in space and time
temporal_jitter	clock noise / timestamp error	time-bin ordering (per polarity)
event_rate_shift	event-density domain gap	global multiplicative rate
polarity_flip	ON/OFF misclassification	polarity sign on a fraction of frames
spatial_dropout	dead-pixel clusters / occlusion	spatial support (rectangular regions zeroed)

Real hardware. We exported the softmax variant to ONNX (static shapes) and measured actual device latency on Qualcomm AI Hub (Qualcomm Technologies, Inc., 2024) (phone-class NPU/CPU) and ST Edge AI Developer Cloud (STMicroelectronics, 2024) (microcontroller board farm; Table 3, main text). Phone devices clear the 50 ms/frame deadline by 1–2 orders of magnitude; of the two bare Cortex-M7 MCUs, one clears it and one misses by $1.048\times$ despite the similar core – board-level factors, not clock speed, decide real-time feasibility without a neural accelerator. Total accuracy cost of the embedded variant: 0.010 mean AUROC ($0.802 \rightarrow 0.792$).

Memory and compute footprint. Both boards’ available internal RAM (455–560 KB) covers the 30.06 KB the model needs by 15–18 $\times$ headroom; the 10.03 MB flash footprint is placed in external flash, which both boards carry (64–128 MB) (numbers per main text, §4.4).

D PHYSICAL MECHANISMS (EXTENDED)

The per-corruption physics summarized in Section 3.2.1, in full; Table 7 maps each corruption to the sensor failure it models and the input axis it perturbs.

Each corruption is grounded in a documented event-camera (DVS) non-ideality, and our histogram-level operation is the faithful tensor-domain analogue of that fault.

- **hot_pixel – defective / stuck pixels, fixed-pattern noise.** Threshold mismatch and leakage make individual pixels fire continuously at fixed coordinates with no scene change: v2e explicitly models such *hot pixels* (Hu et al., 2021), a form of *background-activity* noise that grows in dim/high-gain conditions (Guo & Delbruck, 2023). *Our op:* add a constant count at a fixed set of pixel locations across all frames and time bins: a constant DC offset, the event-camera analogue of stuck CCD/CMOS pixels.
- **event_flood – illumination transients, EMI, readout saturation.** LED/PWM flicker or a strong-light transient drives a dense burst of spurious events; flickering sources “generate many events that provide little or no useful information” (Wang et al., 2022b), and when the local event rate exceeds the readout bandwidth the bus saturates and events are dropped or delayed (Gehrig & Scaramuzza, 2024). *Our op:* inject a localized count burst over a short frame window, bounded in space and time.
- **temporal_jitter – clock noise, timestamp error.** Transistor noise, arbitration/queuing delay, and finite timestamp resolution misassign events to neighbouring time bins; modern row-arbitrated readouts introduce *systematic* timestamp errors (Gehrig & Scaramuzza, 2024; Gallego et al., 2022). *Our op:* roll the ON and OFF time-bin channels by independent integer shifts (zero-filled), preserving the marginal counts up to the edge bins while scrambling the *order* (and hence the temporal autocorrelation) of events.
- **event_rate_shift – contrast-threshold / bias drift, domain gap.** Per-pixel contrast thresholds and bias currents set the event rate, and circuit bias plus manufacturing imperfection make the effective threshold differ across pixels and drift (Wang et al., 2019; Graça et al., 2023); this, or a low- vs. high-texture (slow- vs. fast-motion) domain gap, scales the global rate. *Our op:* multiply all counts by a seeded under- or over-rate factor.
- **polarity_flip – ON/OFF threshold asymmetry.** The ON and OFF thresholds of a pixel are not equal: “warm”/“cool” pixels trigger positive vs. negative events more or less often than expected, an unbalanced-threshold non-ideality (Wang et al., 2019; Hu et al.,

Table 8: Severity schedules. Symbols: n hot pixels, δ injected count, b burst centres, f patch fraction, $s_{\max}$ max bin shift, (u, o) under/over rate factors, p flipped-frame fraction, k dropout regions, (f_h, f_w) region height/width fractions.

Corruption	Sev. 1	Sev. 2	Sev. 3	Sev. 4	Sev. 5
hot_pixel (n, δ)	10, 20	30, 40	80, 80	150, 140	300, 200
event_flood (b, f, δ)	2, .05, 60	4, .08, 100	8, .12, 150	15, .18, 200	25, .25, 255
temporal_jitter $s_{\max}$	1	2	3	5	8
event_rate_shift (u, o)	.80, 1.20	.65, 1.40	.50, 1.65	.35, 2.00	.20, 2.50
polarity_flip p	.05	.10	.20	.35	.50
spatial_dropout (k, f_h, f_w)	1, .05, .05	1, .10, .10	2, .10, .10	2, .15, .15	3, .20, .20

Table 9: Reference detector (Mahalanobis on static φ , full data, L5).

Corruption	Static- φ Mahalanobis AUROC (L5)	Behaviour
hot_pixel	1.000	trivial
temporal_jitter	0.709	moderate
event_rate_shift	0.675	moderate; phase transition $\sim$ sev 3
polarity_flip	0.429	<i>below chance</i> – polarity-symmetric
event_flood	0.408	<i>below chance</i> – moments preserved
spatial_dropout	0.286	<i>anti-detectable</i> ($\rho = -1$)

2021), so the brightness-change sign is misreported, and under fast global illumination swings whole frames can invert. *Our op*: swap the ON/OFF channel blocks on a random fraction of frames.

- **spatial_dropout – dead-pixel clusters, occlusion.** Permanently damaged (dead) pixels, dust/lens contamination, condensation, or partial physical occlusion zero the event support over contiguous areas (Gallego et al., 2022); the related stochastic event-removal primitive is used as a *training* augmentation by EventDrop (Gu et al., 2021). *Our op*: zero one or more rectangular regions across all frames and channels.

E SEVERITY SCHEDULES

The monotone severity parameters for each corruption (referenced from Section 3.2.2).

F BASELINE OOD DETECTORS

We fit seven post-hoc detectors on clean φ and score them on every corrupted run: **Mahalanobis** (Ledit-Wolf; the reference detector), **k -NN** ($k = 5$), **GMM** (5), **OCSVM** (RBF), **PCA** (50), **MLP-AE**, **RealNVP**. Table 9 gives the reference behaviour that frames the benchmark: three corruptions sit *below chance* (informative but inverted – corrupted lies closer to the clean mean than held-out clean).

ANN baselines. A ResNet-18 on event-image / voxel-grid inputs, scored with 16 post-hoc OOD detectors from 2017–2024 (faithful implementations, no retraining) as an upper reference; each carries substantial extra compute (Section C). Full method list and the 6×5 grid ship with the release.

Geometry-aware contraction baseline (k-NNN). k-NNN (Nizan & Tal, 2023), the contraction-geometry baseline closest to the RCF branch’s intent, does *not* help: per-sequence spatial_dropout AUROC is 0.531 versus 0.694 for a plain k -NN – a dedicated method does not beat the simplest neighbour baseline, so the residual is **information-limited, not detector-limited**.

G REPRESENTATION ABLATION

Question. *Is the sub-threshold membrane potential the right signal for OOD detection, or do other features extracted from the same forward pass do better?*

Table 10: **Representation ablation**, per-corruption Mahalanobis AUROC averaged over all five severities; **bold** marks the best representation per corruption. `full_membrane` is φ ; `cls_head` wins on four of six individual corruptions, but φ leads on the overall mean (Figure 3, main text).

Representation	<code>hot_pixel</code>	<code>temp_jitter</code>	<code>rate_shift</code>	<code>polarity</code>	<code>flood</code>	<code>dropout</code>	Mean
Full membrane (φ)	0.963	0.627	0.487	0.470	0.463	0.424	0.572
μ	0.969	0.431	0.516	0.490	0.486	0.449	0.557
σ^2	0.534	0.644	0.478	0.462	0.436	0.423	0.496
κ	0.724	0.673	0.452	0.439	0.418	0.376	0.513
Spike-rate	0.885	0.423	0.517	0.481	0.477	0.441	0.537
Spike-entropy	0.899	0.478	0.607	0.469	0.447	0.413	0.552
ANN features	0.717	0.313	0.483	0.256	0.282	0.273	0.387
Cls-head features	0.647	0.519	0.628	0.505	0.507	0.503	0.552

Setup. We extract eight candidate representations from a single inference pass and score each independently with a one-sided Mahalanobis detector fitted on clean sequences (severity 0). All numbers below are AUROC averaged over all five severities unless noted, averaged over the evaluation set. This section is MDD-free: `full_membrane` is the static φ vector ($[\mu, \sigma^2, \kappa]$ over all four PLIF layers, 2112-D), the same signal the seven classical detectors below use; MDD is evaluated in Section 3.5.

The eight representations are:

- μ, σ^2, κ : per-moment slices of φ (704-D each).
- **full_membrane** (φ): concat of all three moments (2112-D).
- **Spike-rate**: mean firing frequency per PLIF neuron across the sequence.
- **Spike-entropy**: Shannon entropy of the per-neuron firing distribution.
- **ANN features**: GAP of the last ANN neck feature map, i.e. the penultimate representation before the YOLOX detection head.
- **Cls-head features**: GAP of the YOLOX classification sub-head’s first conv output — not final logits (the head has no softmax/sigmoid readout in this probe), but the closest analogue to class-discriminative activations available in a detection model without a softmax classifier.

Results. The pattern is shown in Table 10 (per-corruption breakdown) and Figure 3 (main text, mean over all corruptions), and confirms the theoretical predictions.

ANN features are the weakest overall (Table 10) and collapse hardest on the contractions and polarity: the membrane is not just cheaper than the ANN signal, it is more *informative* for the corruptions that matter most.

Spike-rate struggles at low severity. Averaged over corruptions, spike-rate AUROC is only 0.47 at severity 1 (below chance), rising to 0.56 at severities 3–4, whereas φ starts above chance (0.52) and climbs to 0.59. This is Example 1: the threshold θ gates the spike count, so small sub-threshold shifts leave the firing record untouched while V_{mem} integrates them continuously. Spike-entropy is similarly limited (0.47 at severity 1).

Contraction corruptions. All representations sit near or below 0.5 for `polarity_flip`, `event_flood`, and `spatial_dropout` under one-sided scoring: the contraction inversion of the Section 3.3 taxonomy, which MDD’s two-sided branches address.

Across the seven classical detectors on static φ (mean over 6×5 , Figure 5), PCA-Mahalanobis leads (0.567), followed by Mahalanobis (0.563), GMM (0.551), MLP-AE (0.545), RealNVP (0.542); OCSVM (0.484) and k NN (0.473) fall *below* chance because a one-sided density score inverts on the contraction corruptions. The means cluster near 0.5 precisely because no *single*-sided detector can handle both dilations and contractions: the motivation for MDD.

MDD: from static φ to a full detector. The comparisons above use static φ with a single Mahalanobis pass, the simplest detector on the membrane, at mean AUROC 0.563 (frame). MDD improves on this at every aggregation level, reaching mean AUROC 0.796 at full-sequence aggregation (Table 2, main text, for per-corruption frame/ $W=64$ numbers; Table 14 for the full

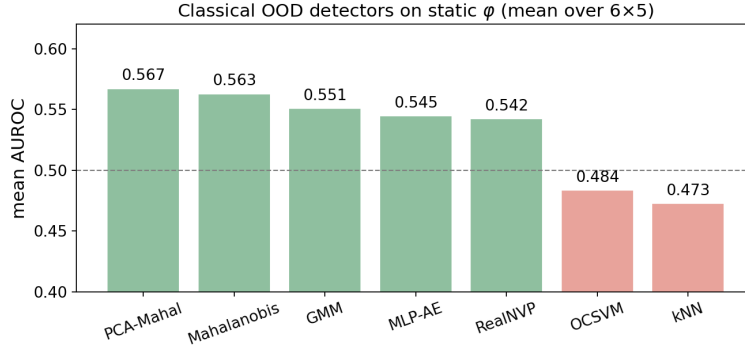


Figure 5: **Seven classical OOD detectors on static ϕ** , mean AUROC over all corruptions/severities. PCA-Mahalanobis/Mahalanobis lead; k NN/OCSVM invert below chance on the contractions: one-sided detectors cannot face both ways, motivating MDD.

Table 11: Per-branch and fused per-sequence AUROC at L5; **fused** is the shipped studentized max -0.5 median.

Corruption	radius	rcf	l4	spatial	fused
hot_pixel	1.000	0.989	1.000	1.000	1.000
event_rate_shift	0.960	0.834	0.506	0.540	0.955
temporal_jitter	0.891	0.394	0.933	0.724	0.952
event_flood	1.000	0.989	0.896	1.000	1.000
polarity_flip	0.613	0.552	0.573	0.639	0.611
spatial_dropout	0.392	0.693	0.403	0.411	0.620

window sweep); per-branch specialisation is in Table 11 (Appendix H). The residual ceiling on polarity_flip/spatial_dropout is discussed in Section 5.

H MDD: PER-BRANCH DECOMPOSITION

Table 11 reports each branch in isolation (per-sequence, L5), with the shipped studentized fusion. Even the fused detector sits *below* the best single branch on both residuals: on polarity_flip, spatial 0.639 > 0.611 fused; on spatial_dropout, rcf 0.693 > 0.620 fused. No combine rule beats the strongest axis on either residual, confirming they are information- rather than fusion-limited (the plain-max vs studentized combiner comparison is Appendix I).

Leave-one-out: every branch is load-bearing. The per-branch table shows what each axis detects *alone*; Table 12 shows what is *lost* when each is removed from the fused max (the complementary “selective fusion” view). The pattern is clean: each branch is the only detector of exactly one corruption, so removing it collapses that corruption while leaving the others untouched. Removing rcf sharply reduces spatial_dropout (0.620 $\rightarrow$ 0.428, -0.192); removing l4 sharply reduces temporal_jitter (0.952 $\rightarrow$ 0.827, -0.125); removing radius costs event_rate_shift (-0.039); and the spatial branch is the marginal protector of the saturated corruptions and polarity_flip (-0.010 to -0.011). No branch is redundant, and no removal helps any corruption by more than $+0.020$ (dropping rcf nudges the dilations up ≤ 0.02 while collapsing spatial_dropout), so a smaller MDD is strictly dominated.

I FUSION-COMBINER ABLATION

To check the studentized max leaves no signal on the table, we benchmarked 16 combiners spanning the outlier-ensemble, copula, rank-aggregation, spectral meta-learning, and standard p -value combiner literatures (full list, faithful reimplementations, and per-combiner AUROC table in the released repository). Only *max*-family combiners stay competitive; every averaging or consensus combiner

Table 12: Leave-one-out branch ablation: fused per-sequence AUROC at L5 with all four branches vs. with each one removed; Δ is the change from full fused (negative = that branch was helping).

Corruption	fused	-radius	-rcf	-l4	-spatial	critical branch
hot_pixel	1.000	1.000	1.000	1.000	0.989	spatial
event_rate_shift	0.955	0.916	0.969	0.948	0.952	radius
temporal_jitter	0.952	0.926	0.972	0.827	0.956	l4
event_flood	1.000	1.000	1.000	1.000	0.989	spatial
polarity_flip	0.611	0.609	0.619	0.611	0.601	spatial
spatial_dropout	0.620	0.603	0.428	0.620	0.625	rcf

MDD is insensitive to k_{pca}/k_{nn} ; farthest-point sampling recovers most of the random-subsample n_{ref} sensitivity at far fewer reference points

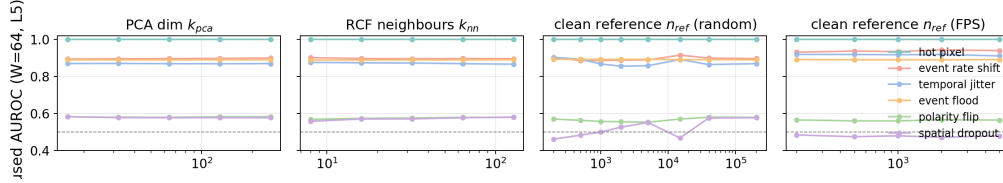


Figure 6: **MDD hyperparameter sensitivity** (fused AUROC, $W=64$, L5): k_{pca} , k_{nn} (flat, log x), a real random-subsample cap on the RCF reference (third panel), and farthest-point sampling replacing it (fourth panel). Discussion above.

(including the spectral meta-learner, the *worst* method) dilutes the residuals, exactly the bias-variance prediction of Aggarwal & Sathe (2015) for outliers flagged by only a few base detectors – consistent with the leave-one-out result (Table 12) that each corruption fires a single branch. Supervised LR fusion and every tested alternative underperform the shipped studentized max -0.5 median; no combine rule recovers the residuals.

J MDD HYPERPARAMETER SENSITIVITY

We sweep each of the three free knobs (k_{pca} (64), RCF k_{nn} (64), clean-reference size n_{ref} (15000)) one at a time, refitting MDD and re-scoring the existing φ (full L5 grid, 168 rows, studentized fusion; Figure 6).

- **k_{pca} , k_{nn} : insensitive.** Fused AUROC varies ≤ 0.006 across $k_{pca} \in [16, 256]$ and ≤ 0.023 across $k_{nn} \in [8, 128]$ (largest swings on the residuals, ≤ 0.010 on the four solved corruptions). No per-dataset tuning needed.
- **n_{ref} (real cap, random subsample): the one sensitive knob.** The shipped knob is itself a no-op (RCF always scores against the full reference); under a *real* cap, mean AUROC ranges 0.783 ($n_{ref}=1000$) to 0.802 (full 204k), almost entirely driven by *spatial_dropout* (0.461 $\rightarrow$ 0.576, the largest swing of any hyperparameter tested); $n_{ref}=5000$ recovers it to 0.551 at 40 $\times$ compression.
- **Farthest-point sampling (FPS) fixes this.** Selecting references by max-min cosine distance instead of randomly keeps mean AUROC flat (0.798 $\rightarrow$ 0.797 across n_{ref}), matching the full-reference mean at 1000 $\times$ fewer stored vectors. Gains are uneven – *temporal_jitter*/*event_rate_shift* improve over random subsampling at matched n_{ref} , but *spatial_dropout* is *not* rescued by FPS either. FPS-200 is the operating point exported and profiled on real hardware (Appendix C).

K SPATIAL-BRANCH MECHANISM CHECK

On a controlled contraction whose signal lives *only* in *phi_spatial* (GAP'd φ matched to clean): fused-without-spatial AUROC = 0.496 $\rightarrow$ fused-with-spatial = 1.000; spatial branch alone = 1.000

Table 13: Effect of the two modifications (AUROC at $W=64$, L5).

Corruption	baseline (max)	radius' + (max - 0.5 med)	+(max - 1.25 med)
hot_pixel	1.000	1.000	1.000
event_rate_shift	0.952	0.956	0.948
temporal_jitter	0.947	0.952	0.966
event_flood	1.000	1.000	1.000
polarity_flip	0.609	0.607	0.560
spatial_dropout	0.560	0.613	0.689
mean	0.845	0.855	0.860
min (worst-corruption floor)	0.560	0.607	0.560

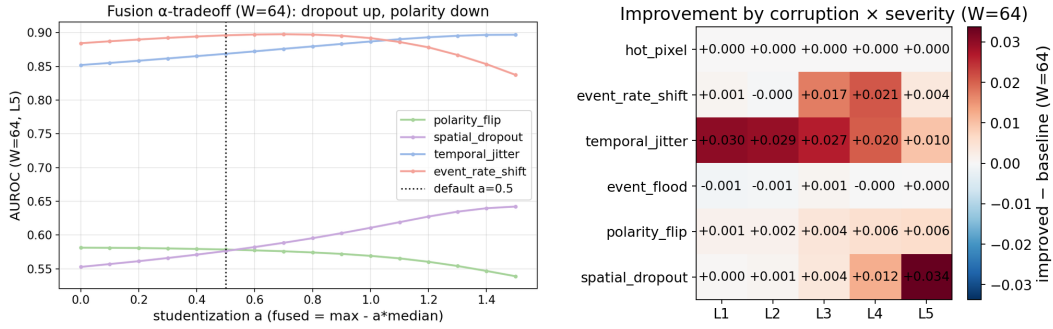


Figure 7: Studentized fusion ($W=64$, L5). (a) α trade-off: increasing α lifts `spatial_dropout` while lowering `polarity_flip` (default $\alpha=0.5$). (b) improved - baseline AUROC per corruption $\times$ severity.

- confirming the wiring. On the banked extraction the spatial branch reaches 1.000 on `event_flood` but stays below chance on `spatial_dropout` (0.411), consistent with the contraction ceiling.

L RESIDUAL-CORRUPTION IMPROVEMENT LOOP

Under a strict *no-re-extraction, no-new-branch* constraint we improved only the existing branches and the fusion rule (mechanics in Section 3.5, main text: `radius'` and the studentized `max -  $\alpha$  · median` combiner). This recovers `spatial_dropout` from 0.576 to 0.620 per sequence with no easy-corruption regression (Table 13, Figure 7b).

Ceiling. Neither residual reaches 0.8: `spatial_dropout` (contraction) caps ≈ 0.69 (rcf branch, Table 11), `polarity_flip` (polarity-symmetric network) ≈ 0.64 (spatial branch, Table 11). Breaking 0.8 needs information not in the current φ (temporal $V(t)$, finer spatial pooling, the raw polarity channel) - out of scope under no-re-extraction.

The full exp1-exp10 trail (including dead ends) is in the released repository.

M SEVERITY AND WINDOW SWEEP (FULL)

The full per-(corruption, severity, branch, window) sweep spans 17 windows. The headline behaviour is `event_flood`'s per-frame $\rightarrow W=64 \rightarrow$ full-recording climb (0.55 $\rightarrow$ 0.89 $\rightarrow$ 1.00) against the two residuals that stay flat at every window (Table 14, Figure 4b in the main text).

Detection-delay vs. accuracy trade-off (`event_flood`). Table 14's `event_flood` row converts directly to a deployment latency at $\Delta t=50$ ms/frame (Figure 8): AUROC crosses the 0.85 operating threshold at $W=64$ (3.2 s decision latency) and reaches 0.976 by $W=128$ (6.4 s); the full-recording ceiling (1.00) requires the whole ~ 730 -frame (36.5 s) recording and is an optimistic upper bound, not a deployable operating point. For a time-critical setting, $W=64$ is the practical knee

Table 14: Fused MDD AUROC vs. aggregation window W (frames) at severity 5, shipped studentized fusion (max -0.5 median); $W=64$ column matches Table 2. $W=1$ is per-frame, “full” is whole-recording; the $W=64$ /full columns carry 95% cluster-bootstrap CIs (resampling recordings, 2000 resamples). A bounded window ($W=32$ – 64) already solves the four detectable corruptions; `spatial_dropout` and `polarity_flip` stay flat (and their CIs straddle or barely exceed 0.5) at *every* window.

Corruption	$W=1$	$W=32$	$W=64$ [95% CI]	$W=128$	full [95% CI]
<code>hot_pixel</code>	1.000	1.000	1.000 [1.000, 1.000]	1.000	1.000 [1.000, 1.000]
<code>event_rate_shift</code>	0.855	0.885	0.896 [0.858, 0.927]	0.914	0.955 [0.914, 0.989]
<code>event_flood</code>	0.551	0.778	0.889 [0.875, 0.904]	0.976	1.000 [1.000, 1.000]
<code>temporal_jitter</code>	0.850	0.859	0.869 [0.836, 0.898]	0.882	0.952 [0.915, 0.984]
<code>spatial_dropout</code>	0.560	0.572	0.576 [0.522, 0.629]	0.584	0.620 [0.537, 0.700]
<code>polarity_flip</code>	0.540	0.573	0.579 [0.522, 0.634]	0.587	0.611 [0.528, 0.690]

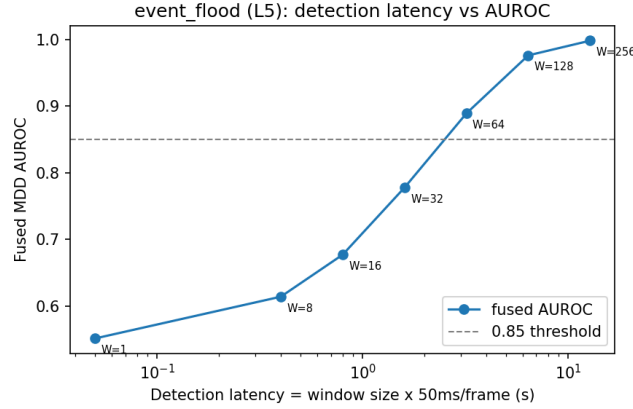


Figure 8: `event_flood` (severity 5): fused MDD AUROC vs. detection latency (window size $\times$ 50 ms/frame, log scale). Same data as the `event_flood` row of Table 14; $W=64$ (3.2 s) is the knee where AUROC first exceeds the 0.85 operating threshold.

of the curve: cutting the window further trades a steep AUROC loss ($0.89 \rightarrow 0.78$ at $W=32$) for a $2\times$ latency win, while widening it past $W=128$ buys little accuracy for a large further latency cost.

Confidence intervals. The $W=64$ /full CIs in Table 14 are a 95% *cluster* bootstrap (recordings resampled with replacement, since frames within a recording are correlated; 2000 resamples).

N SEVERITY MONOTONICITY, CROSS-CORRUPTION, AND WHAT φ ENCODES

Spearman ρ . Score vs. severity: `hot_pixel` +1, `event_flood` +1 (though AUROC barely moves), `polarity_flip` +1, `event_rate_shift` +1, `temporal_jitter` +0.70 ($p = 0.188$, not significant at $n=5$), `spatial_dropout` -1 (perfect anti-detectability).

Cross-corruption zero-shot. We train a single clean-vs-corruption detector on `hot_pixel` only (the one type φ detects perfectly) and evaluate it *zero-shot* on the five held-out corruptions, on validation frames disjoint from training (Figure 10). Transfer essentially fails: every held-out type sits at chance (AUROC ≈ 0.50) regardless of severity, and `event_rate_shift` flips strongly *below* chance, falling to 0.16 at L5 (its activity-driven φ shift points opposite to `hot_pixel`’s). The only weak signal is `event_flood`, which rises monotonically with severity but only to 0.55. So the φ signature is corruption-specific: a detector tuned to one corruption carries essentially no information about the others, which is why MDD fits its branches on clean data alone.

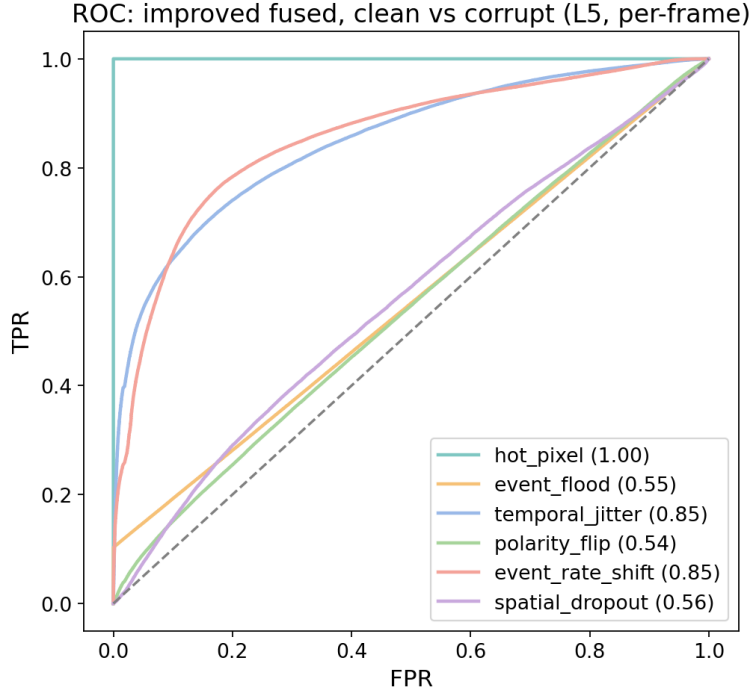


Figure 9: ROC curves of the improved fused MDD at L5 (per-frame), clean vs. each corruption.

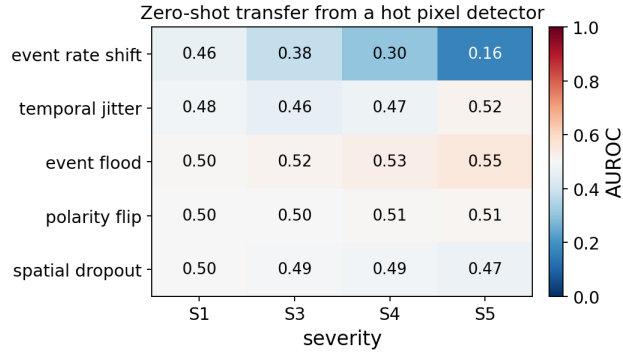


Figure 10: Zero-shot cross-corruption transfer: a clean-vs-hot_pixel logistic detector on the fused φ , evaluated on each held-out corruption (val-only, disjoint from training). Every held-out type sits at or below chance and event_rate_shift inverts to 0.16: the φ signature is corruption-specific, not a generic OOD direction.

O ADDITIONAL GEOMETRY FIGURES

Figure 11 (radial energy and Mahalanobis d^2) and Figure 12 (PCA drift-arrow and mean-shift-cosine views) are both referenced from Section 4.5.

P TEMPORAL REPRESENTATIONS (EXPLORED, NOT USED BY MDD)

For completeness we also explored temporal representations of the membrane trace – handcrafted per-frame statistics of $V(t)$ over the $T=10$ SNN steps (lag-1 autocorrelation, CUSUM, high-frequency energy), a 1D-CNN autoencoder on the pooled trajectory, and margin histograms ($V - \theta$). Under the same leakage-safe split they give only a modest gain over static φ on the hard corruptions

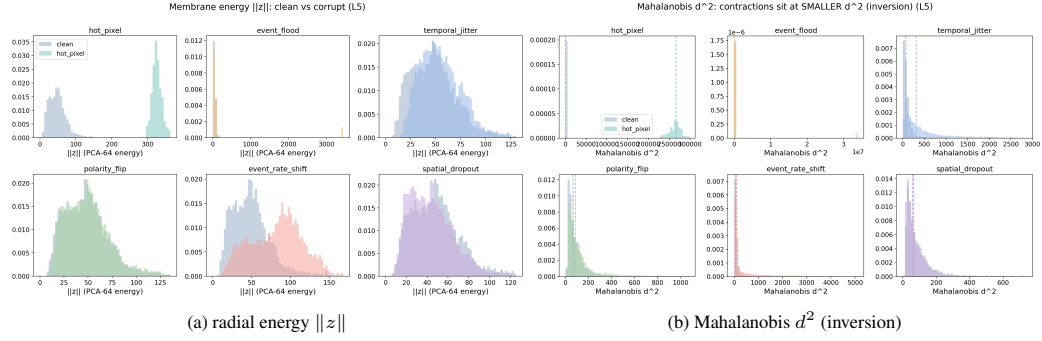


Figure 11: **Geometry of corruption (L5, leakage-safe)**: (a) radial energy $\|z\|$ and (b) Mahalanobis d^2 . PCA drift arrows are below (Figure 12).

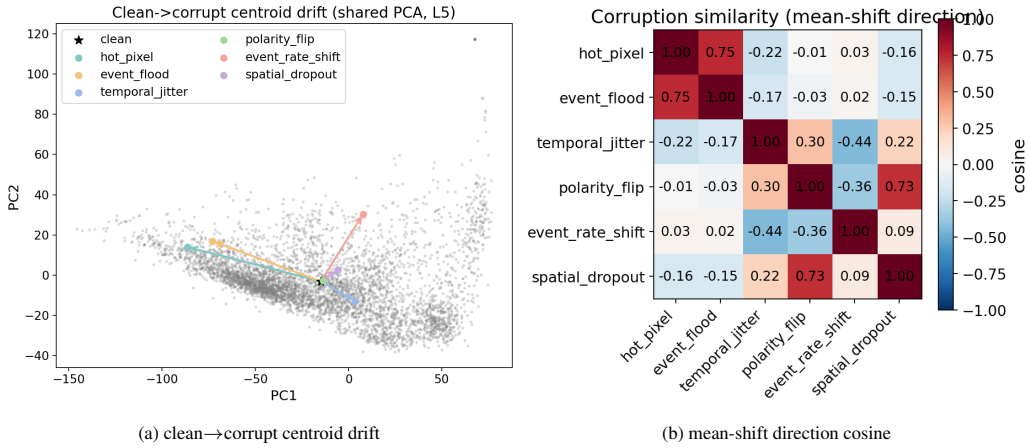


Figure 12: **Corruption-direction geometry (L5)**. (a) centroid drift in a shared clean-PCA; (b) cosine of the mean-shift directions: the two dilations align (hot_pixel $\leftrightarrow$ event_flood, 0.75) as do the two contraction-side residuals (polarity_flip $\leftrightarrow$ spatial_dropout, 0.73), while the taxonomy classes are near-orthogonal across groups.

(~ 0.05 – 0.08 AUROC on event_flood/spatial_dropout), and are *not* part of the deployed MDD, which uses only the static moments $\varphi = [\mu, \sigma^2, \kappa]$.

Q LEVERS TRIED (NEGATIVE-RESULTS CATALOGUE)

Beyond the modifications shipped in Appendix L, we also tried: two-sided/folded Mahalanobis (rescues the below-chance corruptions but degrades temporal_jitter), a global activity scalar (helps only event_rate_shift), and per-feature standardization before Ledoit–Wolf (trades temporal_jitter against event_rate_shift) – each a genuine trade-off rather than a free win, which is why none replaced the shipped default. Full numbers for every lever tried are in the released repository.

R RELEASED ARTIFACTS: GEN1-C

We release the seeded corruption library (regenerating exact streams from the user’s own Gen1 copy, not redistributed histograms), the evaluation harness, and the φ extraction code, so the per-run features (clean + 30 corrupted, with sequence boundaries) can be reproduced without GPU inference from a licensed Gen1 copy. We redistribute no raw Gen1 data; any released derived features follow the Prophesee Gen1 license terms.